Philippe PHILIPPE MARTINET

FRANCE

Directeur.rice de recherche

Poste actuel

03/10/2026 - Présent

Directeur.rice de recherche

Inria

Postes antérieurs

01/01/2017 - 31/12/2022

Directeur.rice de recherche

Inria

01/01/2011 - 31/12/2017

Enseignant.e-chercheur.e/professeur.e

Ecole Centrale de Nantes

01/01/2000 - 31/12/2011

Enseignant.e-chercheur.e/professeur.e

IFMA

01/01/1990 - 31/12/2000

Enseignant.e-chercheur.e/maître.sse de conférence

CUST

01/01/1990 - 31/12/1990

Ingénieur Technico commercial

GST-Alcatel Centre

01/01/1987 - 31/12/1990

Ingénieur (EPIC entreprise)

CSEE

Formations supérieures

01/01/1999 - 31/12/1999

HDR

01/01/1987 - 31/12/1987

PhD

01/01/1985 - 31/12/1985

DEA

Prix, distinction & bourses

03/10/2026

http://www-sop.inria.fr/members/Philippe.Martinet/

03/10/2026

2013: Most active IEEE-RAS TC Award en 2013

03/10/2026

2011: Tan Chin Tuan Exchange Fellowship de l'université de Singapour NTU

03/10/2026

2016: Finalist for the best paper Award at ICARCV16 and EUCOMES16

03/10/2026

2006: CDC06 Best Interactive Paper Award Winners (Best Interactive Session Awards Friday, Dec. 15, 2006)

03/10/2026

2006: CONTROLO06 Best Industrial Paper Award

03/10/2026

2006: ECCV06 Longuet-Higgins Honorable Mention Paper Award

03/10/2026

PEDR 1996-2012 : PES 2012-2016 : PEDR 2016-2020

03/10/2026

2024: Best student paper at ICINCO24

03/10/2026

ANR (VALET, HIANIC, MOBIDEEP, ARMS, FAST, CITYVIP, R-DISCOVER, ARMEN, PROTEUS, COGIRO), FUI (SRDVIAND, VIPA, CRISTAL), Europeen (NEXT, SEAMOVE), ROBEA-CNRS (MAX, MP2, OMNIBOT, R2M, BODEGA), PREDIT Mobivip, RNTL WACIF, Corée (STAR: SAFEMOVE03-04, ICT-ASIA

03/10/2026

Distinctions

Productions scientifiques

Publications

S. Briot, F. Chaumette, P. Martinet, " Revisiting the Determination of the Singularity Cases in the Visual Servoing of Image Points through the Concept of Hidden Robot", in IEEE Transactions on Robotics, Vol. 33, N°2, pp. 536-546, 2017

The hidden robot concept allow to transpose the study of singularities in Sensor based control to the study of equivalent hidden parallel robot.

Publications

P. Long, W. Khalil, P. Martinet, "Dynamic modeling of parallel robots with flexible platforms ", in Mechanism Machine and Theory, 81:21-35, November, 2014

Manipulating deformable object with multi arms system required to model the flexibilities.

Publications

R. Dahmouche, N. Andreff, Y. Mezouar, P. Martinet, O. Ait-Aider " Dynamic visual servoing from sequential regions of interest acquisition", IJRR, Special Issue on Robot Vision, Sage Publications, 31(4): 520-537, April 1st, 2012

Using delays in reading (in CMOS technology) allows to capture velocity information and to design Pose and velocity estimators

Publications

R. Lenain, B. Thuilot, C. Cariou, P. Martinet, "Mixed kinematic and dynamic sideslip angle observer for accurate control of fast off-road mobile robots ", Journal of Field Robotics, 27(2): 181-196, March/April 2010

Taking into account the sliding effect is done. Kinematic and dynamic observers are designed depending on the velocity.

Publications

J. Courbon, Y. Mezouar, P. Martinet, Autonomous navigation of vehicles from a visual memory based on the use of generic camera model, IEEE Trans. ITS, 10(3): 392-402, 2009

A generic camera model has been used in sensory based navigation. A global framework has been established.

Jeux de données, codes sources, logiciels

Liste de tous les papiers y compris les plus récents notamment en IA.

https://cv.hal.science/philippe-martinet

Autres valorisations

2 brevet déposé avec Renault

Autres valorisations

Différent logiciels produits notamment pour l étalonnage de caméra, l'estimation de pose, la création de cartes locales capteurs pour la navigation autonome, l'estimation des paramètres de glissement

Autres valorisations

Beaucoup de robots prototypes mis au point et intégrés (robots mobiles, robots manipulateurs, robots parallèles, UAV, ...)

johann.laconte@inrae.fr

Johann LACONTE

Homme / Man

Autre / Other

Estimation d'état / State Estimation, Navigation autonome / Autonomous Navigation, Cartographie / Mapping

Date de soutenance de thèse / Date of PhD defense

12-2021

Poste(s) actuel(s) / Current position(s)

Code RNSR / R.N.S.R Code	Laboratoire / Laboratory	Code postal / Zip code	Localité / Locality
200118619N	Technologies et systèmes d'information pour les agrosystèmes	63178	AUBIERE

Autres activités / Other activities

- Associate Editor, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- Guest Editor, IEEE Transaction on Field Robotics (T-FR) - Special Issue on ICRA 2025 Workshop on Field Robotics
- Guest Editor, Frontiers in Robotics and AI - AI and Robotics for Smart Agriculture

- Recurring Co-organizer of IEEE International Conference on Robotics and Automation (ICRA) Workshop on Field Robotics (2024,2025,2026)
- Recurring Co-organizer of Forum International de la Robotique Agricole (FIRA) scientific workshop (2024,2025,2026)
- Co-organizer of European Conference on Mobile Robots (ECMR) Workshop on Advanced 3D Vision for Complex Environments (2025)

- Regular reviewer for international journals and conferences (e.g., ICRA, IROS, RA-L)

Responsabilités pédagogiques et administratives / Pedagogical, teaching and administrative responsibilities

- Numerical Signal Processing, ISIMA, Clermont INP
- Mathematical Tools for Robotics, Master Perception Artificielle et Robotique (PAR), UCA
- Control of Robotic Systems, Master Perception Artificielle et Robotique (PAR), UCA
- State Estimation, Master Perception Artificielle et Robotique (PAR), UCA

Poste(s) antérieur(s) / Previous position(s)

De / From	A / To	Nom de l'organisme / Name of the managing authority	Ville / City	Fonction / Function
2022	2023	University of Toronto	Canada	Post-doctorant.e / Post-doctoral Fellow
2022	2022	Université Laval	Canada	Post-doctorant.e / Post-doctoral Fellow
2018	2021	Institut Pascal (UCA)	France	Doctorant.e / PhD Student

Interruption(s) de carrière / Career interruption(s)

Formation supérieure / Higher education

2018-2021 PhD in Robotics, Institut Pascal, Clermont Auvergne University (UCA)
2017–2018 Master Degree in Robotics, Clermont Auvergne University (UCA)
2015–2018 Engineering Degree in Computer Science and Modeling, Institut Supérieur d’Informatique, de Modélisation et de leurs Applications (Clermont Auvergne INP)

Prix, distinctions, bourses / Awards, grants

- Best Paper, IROS 2025 Workshop on Agricultural Robotics and Automation
For the paper "Synthetic Crop-Weed Image Generation and its Impact on Model Generalization", Boyadjian et al., 2025
-Prize Relève étoile Louis-Berlinguet, Field Robotics (FR)
For the paper "Kilometer-scale autonomous navigation in subarctic forests: challenges and lessons learned", Baril et al., 2021
-Second prize of best PhD thesis, GdR Robotique
For the thesis "Lambda-Field : a novel framework for risk assessment in occupancy grids", Johann Laconte, 2021
-Best Robotics Paper, CRV
For the paper “Evaluation of Skid-Steering Kinematic Models for Subarctic Environments”, Baril et al., 2020
-Nomination for Best Student Paper, ICARCV
For the paper "An Information Driven Approach For Ego-Lane Detection Using Lidar And OpenStreetMap", Kasmi et al., 2020

Valorisation / Valorisation

Productions scientifiques / Scientific productions

Publications / Publications

Khizar, Z., Laconte, J., Lenain, R., & Aufrère, R. (2025). Feeling the Force: A Nuanced Physics-based Traversability Sensor for Navigation in Unstructured Vegetation. In 2025 European Conference on Mobile Robots (ECMR)

We developed a novel sensor to measure the resistive forces exerted by vegetation during navigation in tall grass. By capturing these forces, the sensor directly quantifies the energy lost through environmental deformation.

Publications / Publications

Baril, Dominic, et al. "Kilometer-scale autonomous navigation in subarctic forests: challenges and lessons learned." Field Robotics, (2021).

This paper presents an analysis of extensive robotics deployment in northern Canada. It highlights current issues that are yet to solve in robotics, thus providing interesting research avenues.

Publications / Publications

Laconte, Johann, Daniil Lisus, and Timothy D. Barfoot. "Toward Certifying Maps for Safe Registration-Based Localization Under Adverse Conditions." IEEE Robotics and Automation Letters (2023).

We provide theoretical guarantees regarding the ICP algorithm, showing that the resilience to noise is dependent on the environment. Such research is important when deploying robots, as safety and resilience are paramount.

Publications / Publications

Laconte, Johann, et al. "A novel occupancy mapping framework for risk-aware path planning in unstructured environments." Sensors 21.22 (2021): 7562.

This work is the premise of trying to incorporate physical quantities in the robotic navigation stack.

Publications / Publications

Courcelle, Clément, et al. "On the importance of quantifying visibility for autonomous vehicles under extreme precipitation." Towards Human-Vehicle Harmonization 3 (2022): 239.

In this work, we provide an extensive evaluation of the effect of noise on the ICP algorithm, showing that ICP is surprisingly resilient to snowstorm as long as the noise is evenly distributed. Snow gusts, on the contrary, are extremely dangerous for the localization algorithm.

Jeux de données, codes sources, logiciels, data paper, etc. / Data set, software, source code, data paper, etc.

Cropcraft: https://github.com/Romea/cropcraft

CropCraft is a python script that generates 3D models of crop fields, specialized in real-time simulation of robotics applications. - Designed for real-time simulation - Suitable for use with LiDARs and cameras - Highly configurable (YAML file) - Provide ground truth data (identify plant types in LiDAR data)

Jeux de données, codes sources, logiciels, data paper, etc. / Data set, software, source code, data paper, etc.

Natural and Semi-Urban Dataset: https://raoufdannaoui1.github.io/Natural_and_Semi-Urban_Dataset/

A dataset designed to capture short and long-term environmental changes across both natural and semi-urban environments, with specific aim of analyzing how such dynamics impact localization and re-localization performance.

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Mohammad ALKHATIB

Homme / Man

Enseignant.e-chercheur.e/maître.sse de conférence / Lecturer

Intelligence artificielle / Artificial intelligence, Traitement de l'imagerie médicale / Medical imaging processing, Robotique médicale / Medical robotics, Asservissement visuel / Visual servoing, Microrobotique / Microrobotics

Poste(s) actuel(s) / Current position(s)

Code RNSR / R.N.S.R Code	Laboratoire / Laboratory	Code postal / Zip code	Localité / Locality
199412376H	IP	63178	AUBIERE CEDEX

Autres activités / Other activities

Responsabilités pédagogiques et administratives / Pedagogical, teaching and administrative responsibilities

I teach at the University of Clermont Auvergne (at Sigma Engineering School, and at Université Clermont Auvergne) for different student levels (3rd year engineering, 4th year engineering, 5th year engineering, and master level), My courses include Machine Learning, programming for robotics, application of intelligent robotics, Virtual reality for the industry.

Poste(s) antérieur(s) / Previous position(s)

De / From	A / To	Nom de l'organisme / Name of the managing authority	Ville / City	Fonction / Function
2020	2021	Université d'Orléans	Bourges, France	Enseignant.e-chercheur.e/maître.sse de conférence / Lecturer
2017	2020	Université d'Orléans	Bourges, France	Doctorant.e / PhD Student
2012	2015	TOSHIBA Co.	Amman, Jordan	Ingénieur.e (Entreprise) / Engineer (Company)

Interruption(s) de carrière / Career interruption(s)

No period of interruption

Formation supérieure / Higher education

PhD in artificial intelligence and medical robotics 2020

MSc Computer Vision 2017

BSc in Biomedical Engineering 2012

Prix, distinctions, bourses / Awards, grants

Member of IMMORTALLS ANR project

Technical manager of SoftmanBot European project (869855)

Valorisation / Valorisation

Productions scientifiques / Scientific productions

Publications / Publications

Botros, K., Alkhatib, M., Folio, D., & Ferreira, A. (2023). USMicroMagSet: Using deep learning analysis to benchmark the performance of microrobots in ultrasound images. IEEE Robotics and Automation Letters.

A benchmark dataset of microrobots in ultrasound images

Publications / Publications

Alkhatib, M., Hafiane, A., & Vieyres, P. (2021, January). Merged 1D-2D deep convolutional neural networks for nerve detection in ultrasound images. In 2020 25th International Conference on Pattern Recognition (ICPR) (pp. 4774-4780). IEEE.

1- Automatic detection of nerve structure in ultrasound images. 2- Proposing a robust merged CNNs model based on texture analysis and CNN models.

Publications / Publications

Alkhatib, M., Hafiane, A., Vieyres, P., & Delbos, A. (2019). Deep visual nerve tracking in ultrasound images. Computerized Medical Imaging and Graphics, 76, 101639.

1- Tracking of nerve structure in ultrasound images. 2- Comparative study of recent deep-learning tracking techniques.

Publications / Publications

Robust adaptive median binary pattern for noisy texture classification and retrieval

1- Robust descriptor for different high noises. 2- High noisy texture classification. 3- Noise-free texture classification. 4- Noisy texture retrieval.

Publications / Publications

Botros, K., Alkhatib, M., Folio, D., & Ferreira, A. (2022, May). Fully automatic and real-time microrobot detection and tracking based on ultrasound imaging using deep learning. In 2022 International Conference on Robotics and Automation (ICRA) (pp. 9763-9768). IEEE.

Deep learning techniques for automatic and real-time detection and tracking of microrobots in ultrasound images

Jeux de données, codes sources, logiciels, data paper, etc. / Data set, software, source code, data paper, etc.

The USMicroMagset dataset is available on gitlab at https://gitlab.com/insacvl2/ USMicroMagSet/

The first publicly available B-mode ultrasound dataset for microrobots (USmicroMagSet) with accurate annotations which contains more than 40000 samples of magnetic microrobots.